

Statistical Reinforcement Learning in Three Acts

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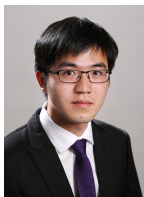
My amazing co-conspirators



Jiin Woo
CMU



Sudeep Salgia
CMU



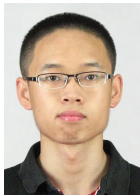
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Gen Li
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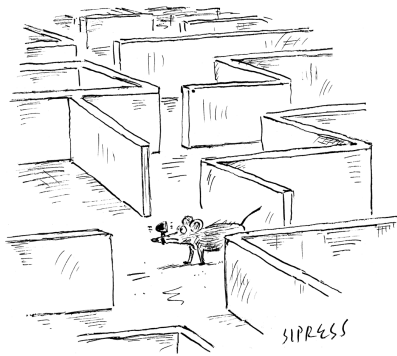
Yuxin Chen
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Reinforcement learning (RL)

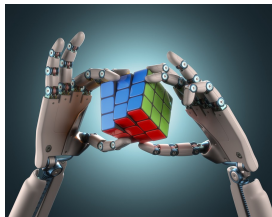
In RL, an agent learns by interacting with an *unknown* environment through trial-and-error to maximize long-term total reward.



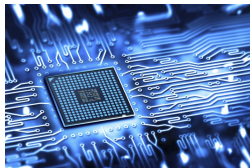
"Recalculating ... recalculating ..."



More successes of RL since AlphaGo



robotics



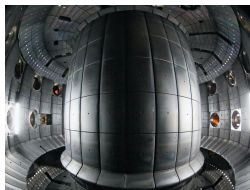
chip designs



resource management



strategic games

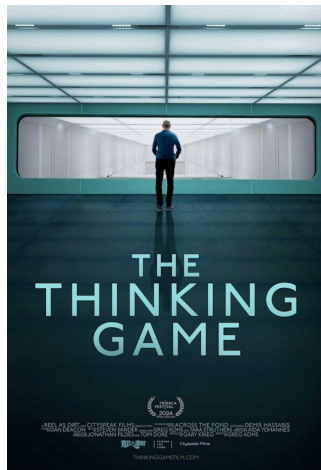


nuclear plant control



UAV and drones

One more: RL for thinking models



Turing Award Goes to 2 Pioneers of Artificial Intelligence

Andrew Barto and Richard Sutton developed reinforcement learning, a technique vital to chatbots like ChatGPT.



RL holds great promise in accelerating scientific, engineering and societal discoveries.

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How do we build the statistical and algorithmic foundations of RL to help realizing its potential?

Sample efficiency

However, collecting data samples might be expensive or time-consuming.



clinical trials

Prompt:

Should I add chorizo
to my paella?

Response 1: Absolutely! ...

Response 2: In Valencian...

Feedback (ranking):

Response 1 is better than 2

LLM alignment



autonomous driving

Sample efficiency

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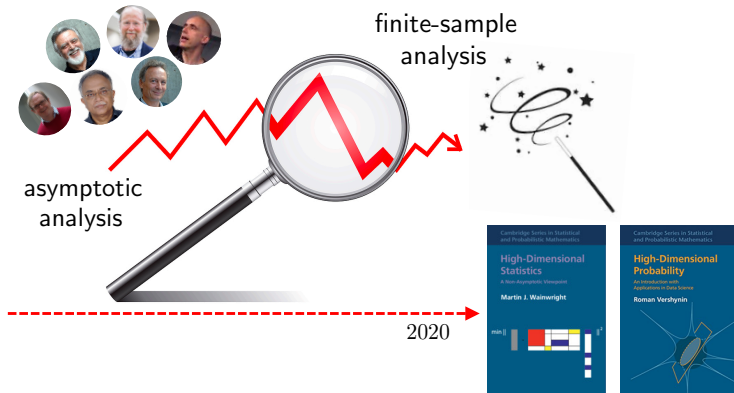
autonomous driving

Calls for design of sample-efficient RL algorithms!

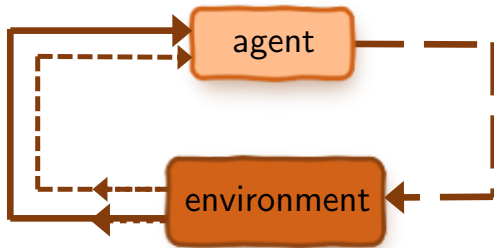


asymptotic
analysis





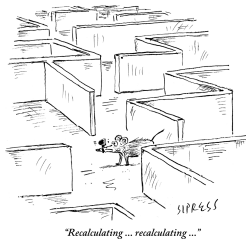
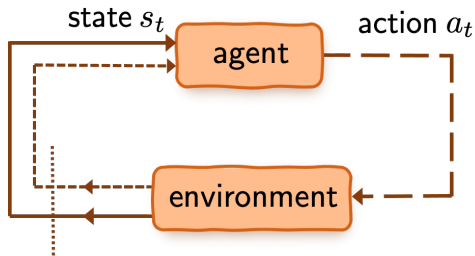
Non-asymptotic analyses are key to understand and improve statistical efficiency in modern RL.



The prologue: Markov decision processes

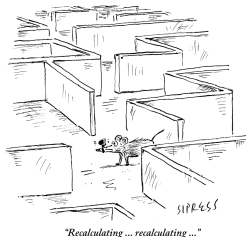
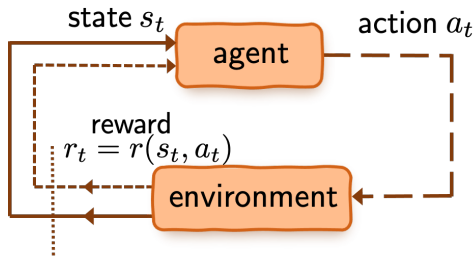


Markov decision process (MDP)



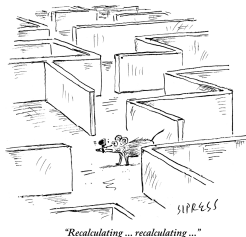
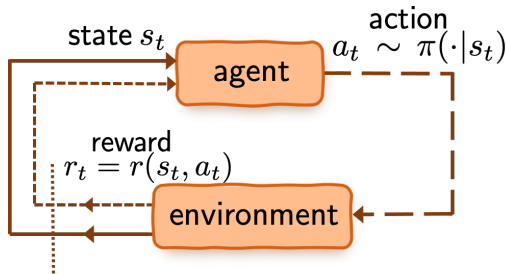
- $\mathcal{S}$: state space
- $\mathcal{A}$: action space

Markov decision process (MDP)



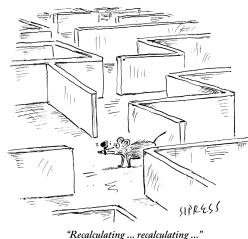
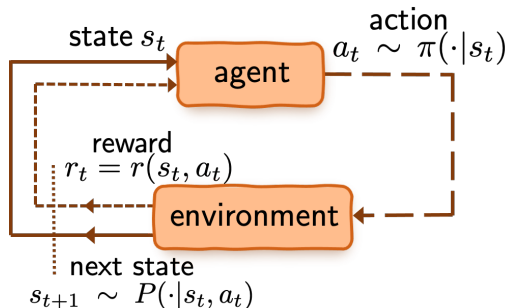
- $\mathcal{S}$: state space
- $\mathcal{A}$: action space
- $r(s, a) \in [0, 1]$: immediate reward

Markov decision process (MDP)



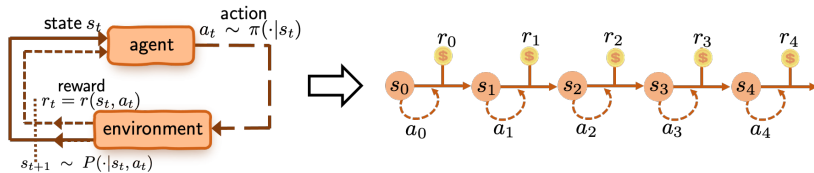
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- $\pi(\cdot | s)$: policy (or action selection rule)

Markov decision process (MDP)

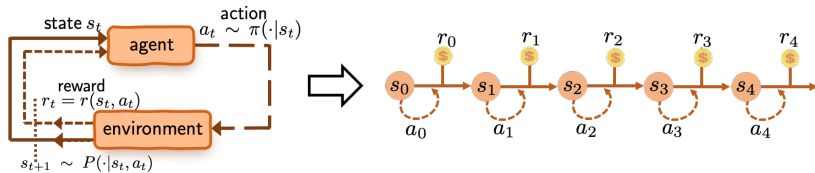


- $\mathcal{S}$: state space
- $\mathcal{A}$: action space
- $r(s, a) \in [0, 1]$: immediate reward
- $\pi(\cdot|s)$: policy (or action selection rule)
- $P(\cdot|s, a)$: transition probabilities

Value function



Value function



Value function of policy π :

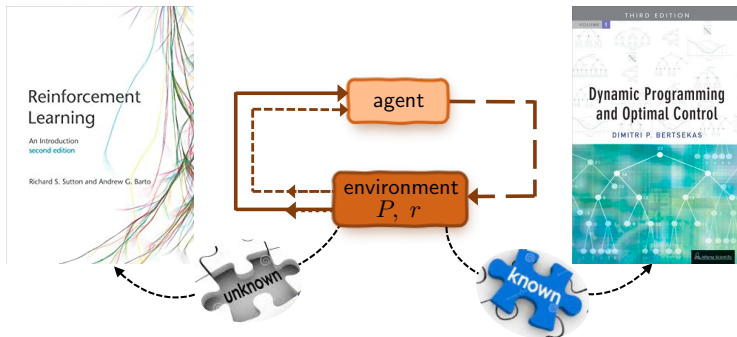
$$\forall s \in \mathcal{S}: \quad V^\pi(s) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right]$$

Q-function of policy π :

$$\forall (s, a) \in \mathcal{S} \times \mathcal{A}: \quad Q^\pi(s, a) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_0 = a \right]$$

- $\gamma \in [0, 1)$ is the **discount factor**; $\frac{1}{1-\gamma}$ is **effective horizon**
- Expectation is w.r.t. the sampled trajectory under π

Searching for the optimal policy



Goal: find the optimal policy π^* that maximize $V^\pi(s)$

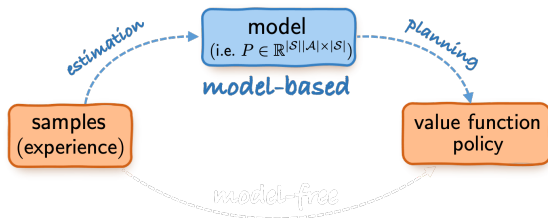
- optimal value / Q function: $V^* := V^{\pi^*}$, $Q^* := Q^{\pi^*}$
- optimal policy $\pi^*(s) = \operatorname{argmax}_{a \in \mathcal{A}} Q^*(s, a)$

Central question: *how many samples are sufficient to learn an ε -optimal policy?*

- **Act 1:** what is the model good for?
- **Act 2:** the statistical price of robustness
- **Act 3:** the blessing of heterogeneity of going federated

Act 1: what is the model good for?

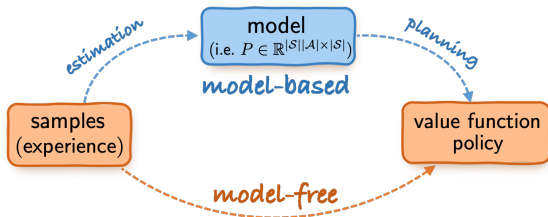
Algorithmic primitives



Model-based approach (“plug-in”)

1. build an empirical estimate $\hat{P}$ for P
2. planning based on the empirical $\hat{P}$

Algorithmic primitives



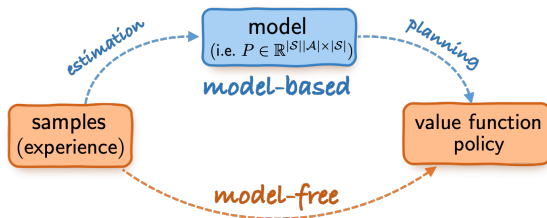
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Model-free approach

— learning w/o estimating the model explicitly

Algorithmic primitives



Model-based approach (“plug-in”)

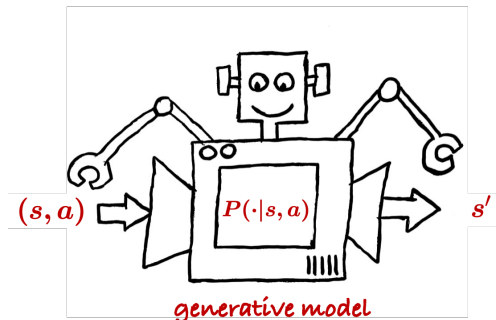
1. build an empirical estimate $\hat{P}$ for P
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Model-free approach

— learning w/o estimating the model explicitly

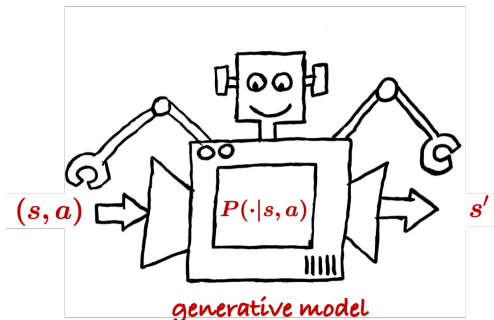
Is there a statistical gap between them?

Model estimation under the generative model



For each (s, a) , collect N ind. samples $\{(s, a, s'_{(i)})\}_{1 \leq i \leq N}$

Model estimation under the generative model

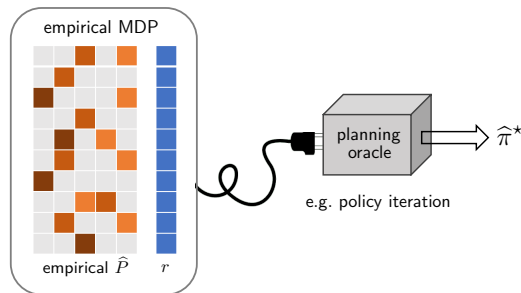


For each (s, a) , collect N ind. samples $\{(s, a, s'_{(i)})\}_{1 \leq i \leq N}$

Empirical estimates: estimate $\hat{P}(s'|s, a)$ by $\underbrace{\frac{1}{N} \sum_{i=1}^N \mathbb{1}\{s'_{(i)} = s'\}}_{\text{empirical frequency}}$

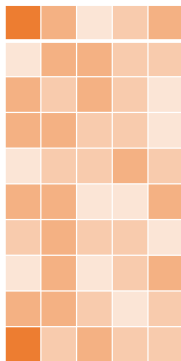
Model-based (plug-in) estimator

(Azar et al., 2013; Agarwal et al., 2019)

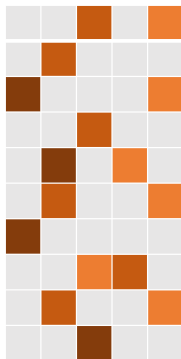


Run planning algorithms based on the *empirical* MDP

Challenges in the sample-starved regime



truth: $P \in \mathbb{R}^{SA \times S}$



empirical $\hat{P}$ (sparse)

- can't recover P faithfully if sample size $\ll S^2 A$

Can we trust the policy estimate when reliable model est. is infeasible?

Minimax optimality of the plug-in estimator

Theorem (Li et al., OR 2024)

For any $0 < \varepsilon \leq \frac{1}{1-\gamma}$, the plug-in estimator finds a policy $\hat{\pi}$ that satisfies $V^ - V^{\hat{\pi}} \leq \varepsilon$ with sample complexity at most*

$$\tilde{O}\left(\frac{|S||\mathcal{A}|}{(1-\gamma)^3\varepsilon^2}\right).$$

Minimax optimality of the plug-in estimator

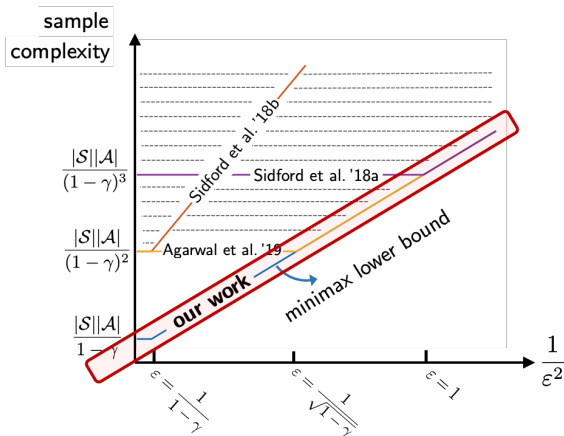
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For any $0 < \varepsilon \leq \frac{1}{1-\gamma}$, the plug-in estimator finds a policy $\hat{\pi}$ that satisfies $V^* - V^{\hat{\pi}} \leq \varepsilon$ with sample complexity at most

$$\tilde{O}\left(\frac{|S||\mathcal{A}|}{(1-\gamma)^3\varepsilon^2}\right).$$

- Matches with **minimax lower bound**: $\tilde{\Omega}\left(\frac{SA}{(1-\gamma)^3\varepsilon^2}\right)$ (Azar et al. '13)
- We can find the optimal policy without reliable model estimation!

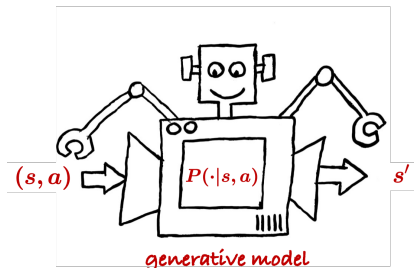
The plug-in estimator is minimax-optimal



Indeed, the plug-in approach is also minimax-optimal for:

- offline RL (Li et al. '24)
- online RL (Zhang et al. '24)

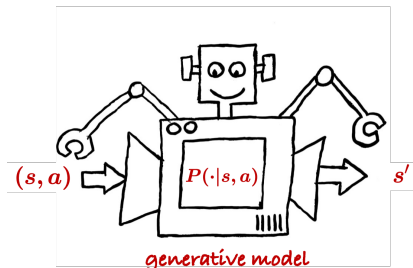
Going model-free: Q-learning



Stochastic approximation for solving Bellman equation $Q^* = \mathcal{T}(Q^*)$ using samples collected from the generative model:

$$Q_{t+1}(s, a) = (1 - \eta)Q_t(s, a) + \eta \underbrace{\mathcal{T}_t(Q_t)(s, a)}_{\text{draw the transition } (s, a, s') \text{ for all } (s, a)}, \quad t \geq 0$$

Going model-free: Q-learning



Stochastic approximation for solving Bellman equation $Q^* = \mathcal{T}(Q^*)$ using samples collected from the generative model:

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$$\mathcal{T}_t(Q)(s, a) = r(s, a) + \gamma \max_{a'} Q(s', a')$$

$$\mathcal{T}(Q)(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} \left[\max_{a'} Q(s', a') \right]$$

Sample complexity of Q-learning

Theorem (Li et. al., OR 2024)

For any $0 < \varepsilon \leq 1$, Q-learning yields $\|\widehat{Q} - Q^*\|_\infty \leq \varepsilon$ with sample complexity *at most*

$$\tilde{O}\left(\frac{|\mathcal{S}||\mathcal{A}|}{(1-\gamma)^4\varepsilon^2}\right).$$

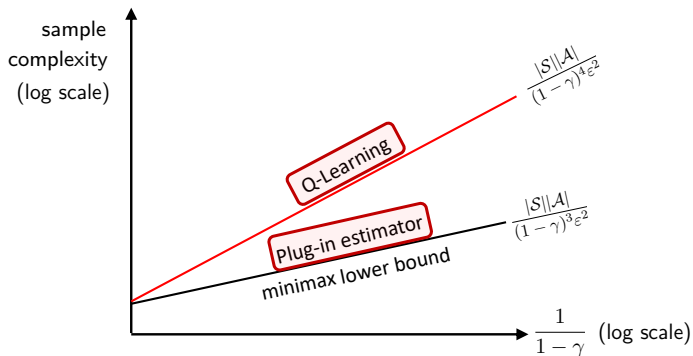
Furthermore, there exists some MDP such that this requirement is tight.

- Allows both constant and rescaled linear learning rate:

$$\frac{1}{1 + \frac{c_1(1-\gamma)T}{\log^2 T}} \leq \eta_t \leq \frac{1}{1 + \frac{c_2(1-\gamma)t}{\log^2 T}}$$

- Deeply related to the “over estimation” issue (Thrun and Schwartz, 1993; Hasselt, 2010): $\max_{a \in \mathcal{A}} \mathbb{E}X(a)$ tends to be over-estimated (high positive bias) when $\mathbb{E}X(a)$ is replaced by its empirical estimates.

Where we stand



A sample complexity separation
between model-based and model-free approaches

Act 2: the statistical price of robustness

Safety and robustness in RL

(Zhou et al., 2021; Panaganti and Kalathil, 2022; Yang et al., 2022;)



Training environment

≠



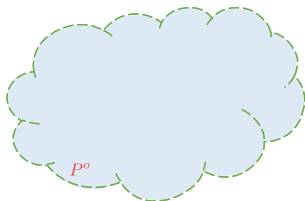
Test environment

Sim2Real Gap: Can we learn optimal policies that are robust to model perturbations?

Distributionally robust MDP

Uncertainty set of the nominal transition kernel P^o :

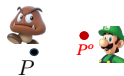
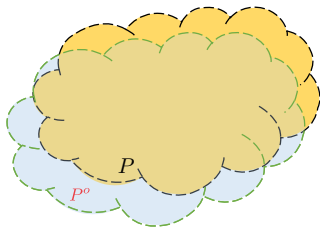
$$\mathcal{U}^\sigma(P^o) = \{P : \rho(P, P^o) \leq \sigma\}, \quad \text{where } \rho \text{ is some distance metric}$$



Distributionally robust MDP

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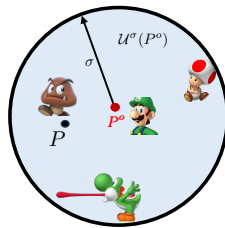
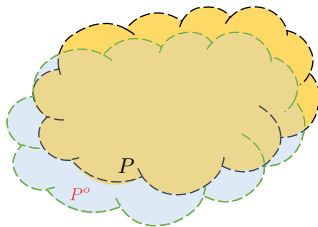
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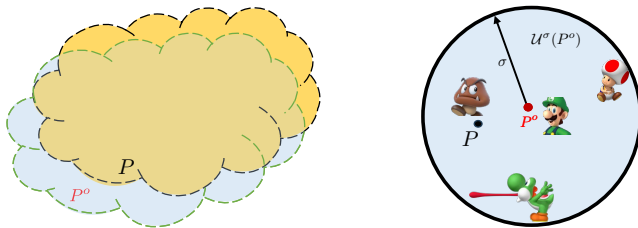
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Distributionally robust MDP

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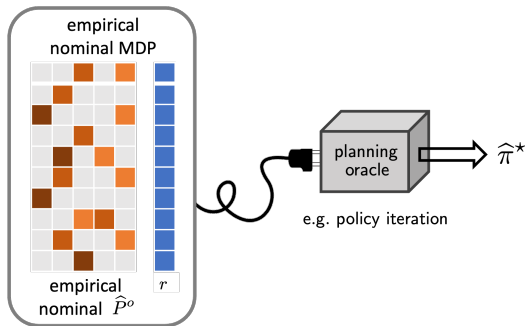
Find the policy π^* that maximizes the **robust value** of policy π :

$$\forall s \in \mathcal{S}: \quad V^{\pi, \sigma}(s) := \inf_{P \in \mathcal{U}^\sigma(P^o)} \mathbb{E}_{\pi, P} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right]$$

Measures the **worst-case** performance of the policy in the uncertainty set.

Model-based planning in robust MDP

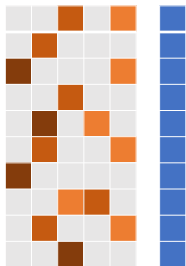
(Iyengar. '05, Nilim and El Ghaoui. '05)



Planning by **distributionally robust value iteration (DRVI)**:

$$\widehat{Q}(s, a) \leftarrow r(s, a) + \gamma \inf_{P_{s,a} \in \mathcal{U}^\sigma(\hat{P}_{s,a}^o)} \langle P_{s,a}, \widehat{V} \rangle, \quad \widehat{V}(s) = \max_a \widehat{Q}(s, a).$$

A curious question



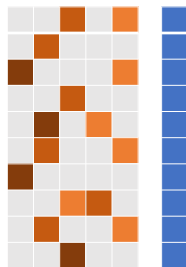
empirical MDP

Learn the optimal policy of
the nominal MDP?

Learn the **robust** policy
around the nominal MDP?



A curious question



empirical MDP

Learn the optimal policy of the nominal MDP?

Learn the **robust** policy around the nominal MDP?



Robustness-statistical trade-off? Is there a statistical premium that one needs to pay in quest of additional robustness?

Sample complexity under TV uncertainty

Theorem (Shi et al., 2023)

Assume the uncertainty set is measured via the *TV distance* with radius $\sigma \in [0, 1)$. For sufficiently small $\varepsilon > 0$, DRVI outputs a policy $\hat{\pi}$ that satisfies $V^{*,\sigma} - V^{\hat{\pi},\sigma} \leq \varepsilon$ with sample complexity at most

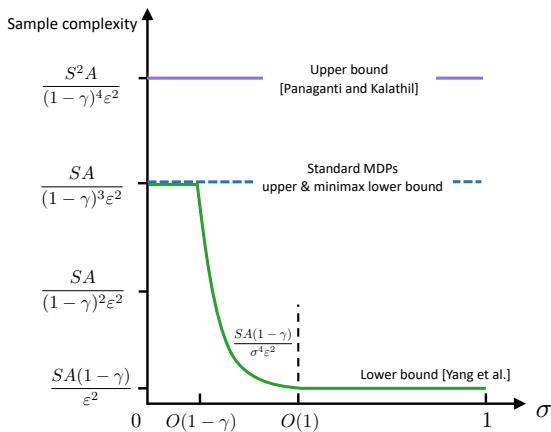
$$\tilde{O}\left(\frac{SA}{(1-\gamma)^2 \max\{1-\gamma, \sigma\} \varepsilon^2}\right)$$

ignoring logarithmic factors. In addition, no algorithm can succeed if the sample size is below

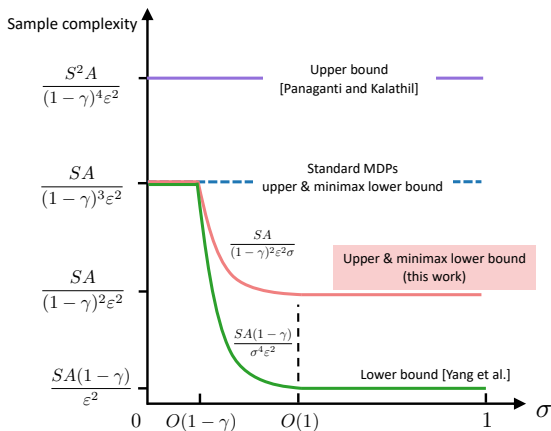
$$\tilde{\Omega}\left(\frac{SA}{(1-\gamma)^2 \max\{1-\gamma, \sigma\} \varepsilon^2}\right).$$

- Establish the minimax optimality of DRVI for RMDP under the TV uncertainty set over the full range of σ .

When the uncertainty set is TV



When the uncertainty set is TV



RMDPs are **easier** to learn than standard MDPs.

Sample complexity under χ^2 uncertainty

Theorem (Shi et al., 2025)

Assume the uncertainty set is measured via the χ^2 divergence with radius $\sigma \in [0, \infty)$. For sufficiently small $\varepsilon > 0$, DRVI outputs a policy $\hat{\pi}$ that satisfies $V^{*,\sigma} - V^{\hat{\pi},\sigma} \leq \varepsilon$ with sample complexity at most

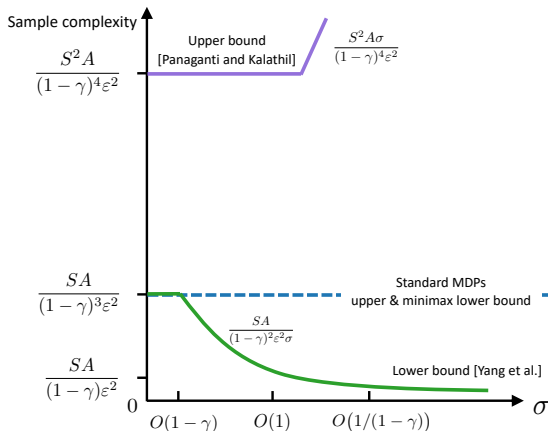
$$\tilde{O}\left(\frac{SA}{(1-\gamma)^3\varepsilon^2}\left(1 + \frac{\sqrt{\sigma} + \sigma}{1-\gamma}\right)\right)$$

ignoring logarithmic factors. In addition, no algorithm can succeed if the sample size is below

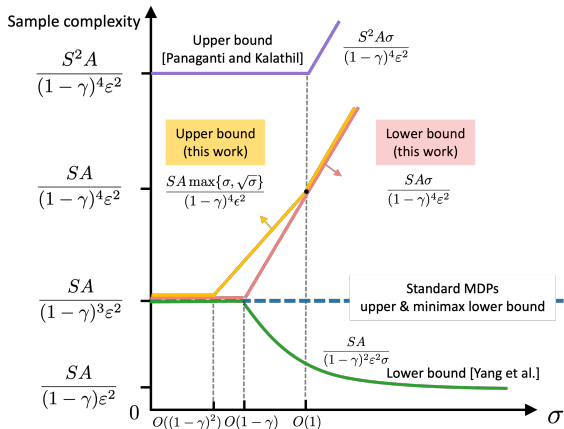
$$\tilde{\Omega}\left(\frac{SA}{(1-\gamma)^3\varepsilon^2}\left(1 + \frac{\sigma}{1-\gamma}\right)\right)$$

- Establish the minimax optimality of DRVI for RMDP under the TV uncertainty set over the full range of σ .

When the uncertainty set is χ^2 divergence

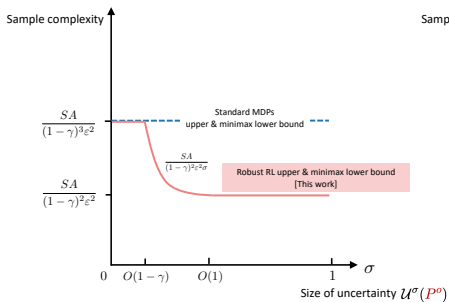


When the uncertainty set is χ^2 divergence

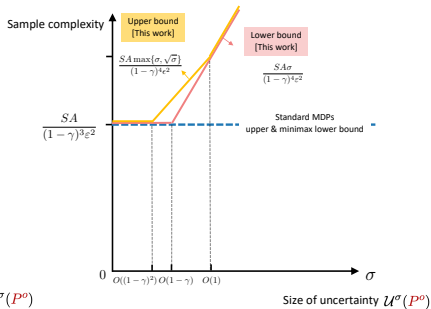


RMDPs are **harder** to learn than standard MDPs.

Where we stand

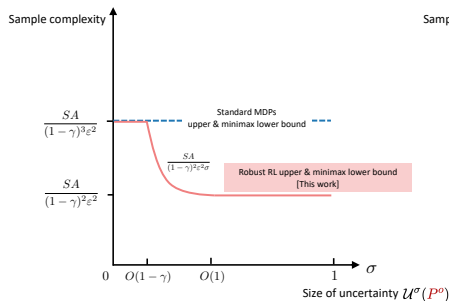


(a) TV distance: Easier than standard RL

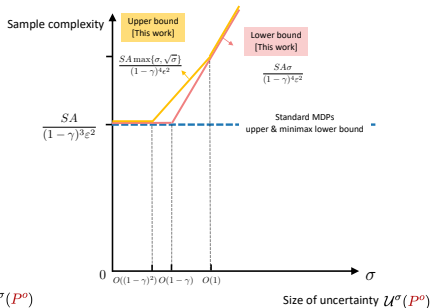


(b) χ^2 divergence: Harder than standard RL

Where we stand



(a) TV distance: Easier than standard RL

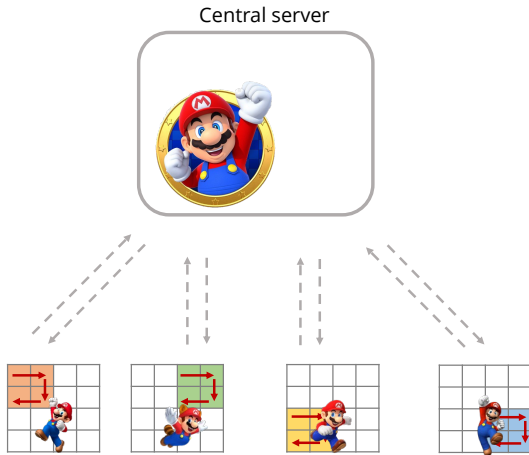


(b) χ^2 divergence: Harder than standard RL

The statistical price of robustness:
sometimes it's cheap, sometimes it's not!

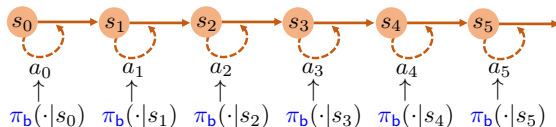
Act 3: the blessing of heterogeneity in federated RL

RL meets federated learning



Federated reinforcement learning: enables multiple agents to collaboratively learn a global policy without sharing datasets.

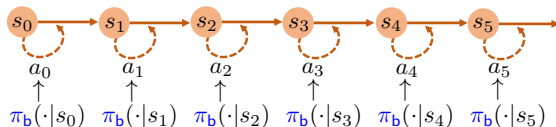
Q-learning following a behavior policy



Stochastic approximation for solving Bellman equation $Q^* = \mathcal{T}(Q^*)$ using samples collected from a behavior policy π_b :

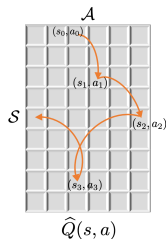
$$Q_{t+1}(s_t, a_t) = \underbrace{(1 - \eta)Q_t(s_t, a_t) + \eta \mathcal{T}_t(Q_t)(s_t, a_t)}_{\text{only update } (s_t, a_t)\text{-th entry}}, \quad t \geq 0$$

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$$\mathcal{T}_t(Q)(s_t, a_t) = r(s_t, a_t) + \gamma \max_{a'} Q(s_{t+1}, a')$$

$$\mathcal{T}(Q)(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a'} Q(s', a') \right]$$

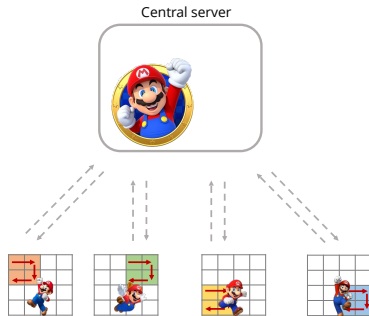
Federated Q-learning with local updates

Alternates between local computation and global communication

- **Local updates:** the agent k performs τ rounds of local Q-learning updates:

$$Q_{t+1}^k \leftarrow (1 - \eta)Q_t^k + \eta \mathcal{T}_t(Q_t^k)$$

and sends it to the server.



Federated Q-learning with local updates

Alternates between local computation and global communication

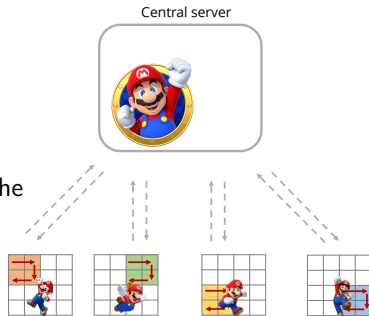
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and sends it to the server.

- **Periodic averaging:** the server averages the local updates and sends it back to agents:

$$Q_t = \frac{1}{K} \sum_{k=1}^K Q_t^k$$

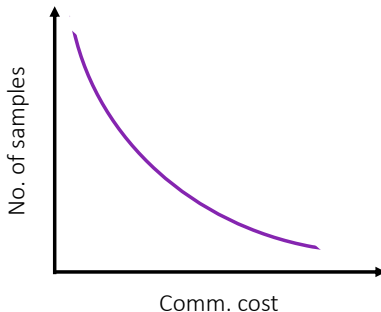


Statistical-communication trade-offs

Statistical
benefits



Communication
overhead



What is the benefit of collaboration, and is there a price?

A communication lower bound for federated Q-learning

Theorem (Salgia and Chi, NeurIPS 2024; informal)

For a wide family of federated Q-learning algorithm with intermittent communication, regardless of the choice of synchronization schedules, the number of communication rounds needs to be at least

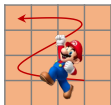
$$\tilde{\Omega}\left(\frac{1}{1-\gamma}\right)$$

in order to achieve any speedup with respect to the number of agents.

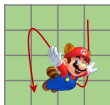
- A similar lower bound holds for the number of communication *bits*.
- Bias-variance trade-off: **bias** increases between two communication rounds, while **variance** exhibits linear speedup on average.
- Note: one-shot average suffices for federated TD learning.

The benefit of collaboration?

Single-agent RL requires **full coverage** of the agent over the entire state-action space...

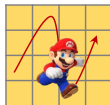


Agent 1



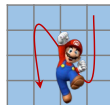
Agent 2

...



Agent k

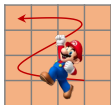
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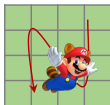
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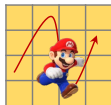


Agent 1



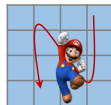
Agent 2

...



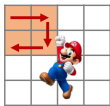
Agent k

...

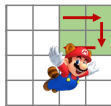


Agent K

However, the power of collaboration really shines if we only need...

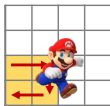


Agent 1



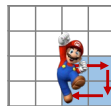
Agent 2

...



Agent k

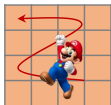
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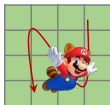
Agent K

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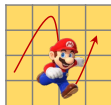


Agent 1

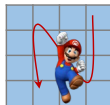


Agent 2

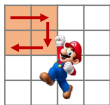
...

Agent k

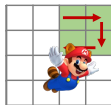
...

Agent K

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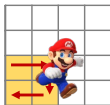


Agent 1

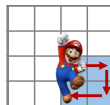


Agent 2

...

Agent k

...

Agent K

Is collaborative coverage enough for federated Q-learning?

Key metrics

Collaborative coverage: minimum entry of the average stationary distribution

$$\mu_{\text{avg}} = \min_{s,a} \frac{1}{K} \sum_{k=1}^K \mu_{\text{b}}^k(s, a).$$

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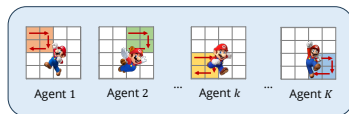
$$\mu_{\text{avg}} = \min_{s,a} \frac{1}{K} \sum_{k=1}^K \mu_{\text{b}}^k(s,a).$$

Heterogeneity of local behavior policies: density ratio of individual / average behavior policies

$$C_{\text{het}} = K \max_{k,s,a} \frac{\mu_{\text{b}}^k(s,a)}{\sum_{k=1}^K \mu_{\text{b}}^k(s,a)} = \max_{k,s,a} \frac{\mu_{\text{b}}^k(s,a)}{\mu_{\text{avg}}(s,a)}.$$



$$C_{\text{het}} = 1$$



$$C_{\text{het}} = K$$

Our theorem

Theorem (Woo, Joshi, Chi, JMLR 2025)

For sufficiently small $\varepsilon > 0$, federated asynchronous Q-learning yields $\|\widehat{Q} - Q^\|_\infty \leq \varepsilon$ with sample complexity at most*

$$\tilde{O}\left(\frac{C_{\text{het}}}{K \mu_{\text{avg}} (1 - \gamma)^5 \varepsilon^2}\right)$$

ignoring the burn-in cost that depends on the mixing times.

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- Collaborative coverage suffices!
- Near-optimal **linear speedup** when the local behavior policies are similar, $C_{\text{het}} \approx 1$.

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Curse of heterogeneity? Performance degenerates when local behavior policies are heterogeneous (i.e. $1 \ll C_{\text{het}}$). ☹

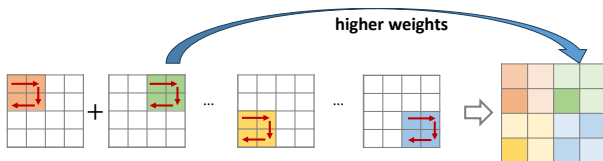
Importance averaging

Key observation: not all updates are of same quality due to limited visits induced by the behavior policy.



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Importance averaging: the server averages the local updates based on importance via

$$Q_t(s, a) = \frac{1}{K} \sum_{k=1}^K \alpha_t^k(s, a) Q_t^k(s, a),$$

where

$$\alpha_t^k = \frac{(1 - \eta)^{-N_{t-\tau, t}^k(s, a)}}{\sum_{k=1}^K (1 - \eta)^{-N_{t-\tau, t}^k(s, a)}}, \quad N_{t-\tau, t}^k(s, a) = \begin{array}{l} \text{number of visits} \\ \text{in the sync period} \end{array}.$$

The benefit of heterogeneity

Theorem (Woo, Joshi, Chi, JMLR 2025)

For sufficiently small $\varepsilon > 0$, federated asynchronous Q-learning *with importance averaging* yields $\|\widehat{Q} - Q^*\|_\infty \leq \varepsilon$ with at most

$$\tilde{O}\left(\frac{1}{K\mu_{\text{avg}}(1-\gamma)^5\varepsilon^2}\right)$$

samples, ignoring the burn-in cost that depends on the mixing times.

The benefit of heterogeneity

Theorem (Woo, Joshi, Chi, JMLR 2025)

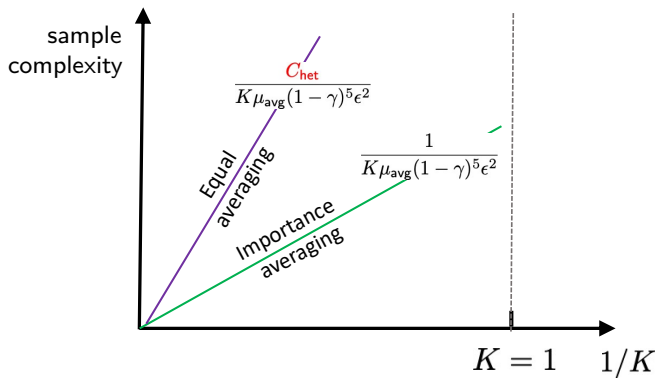
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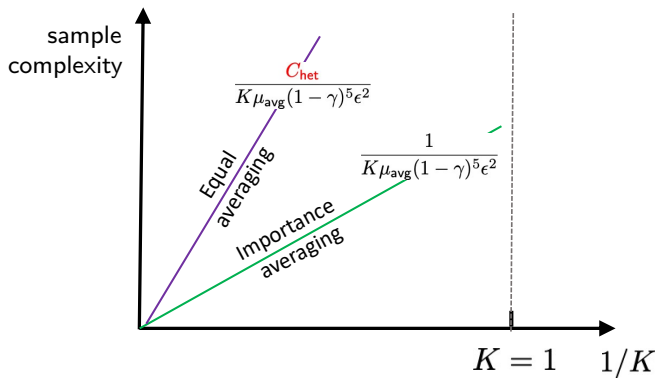
samples, ignoring the burn-in cost that depends on the mixing times.

- Similar results can be developed for the offline setting, too.
- This is comparable to the centralized performance when we colocate all the data.

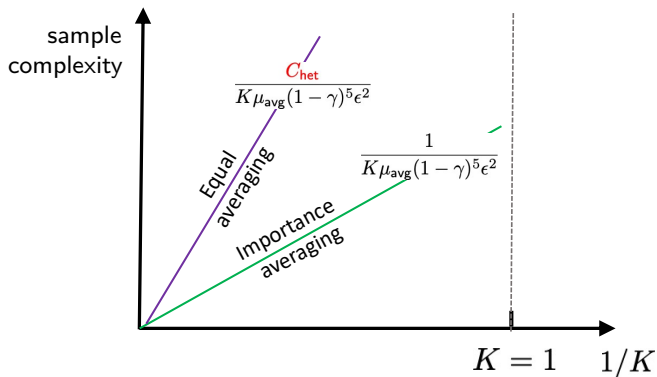
Where we stand



Where we stand

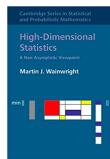
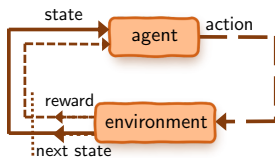
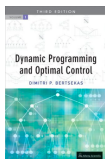
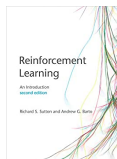


Where we stand



Federated Q-learning harnesses data heterogeneity!

Epilogue



Understanding non-asymptotic performances of RL algorithms is a fruitful playground to unveil and tackle algorithmic challenges!

What we have covered: the separation of model-based and model-free RL, statistical price of robustness, and federated RL to embrace heterogeneity.

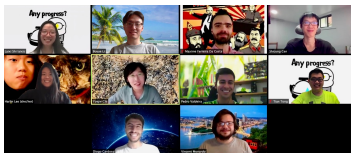
What we have not covered:

- function approximation
- multi-agent RL
- RL for foundation models
- many more...

Thanks!



The χ Group



<https://yuejiechi.github.io/>